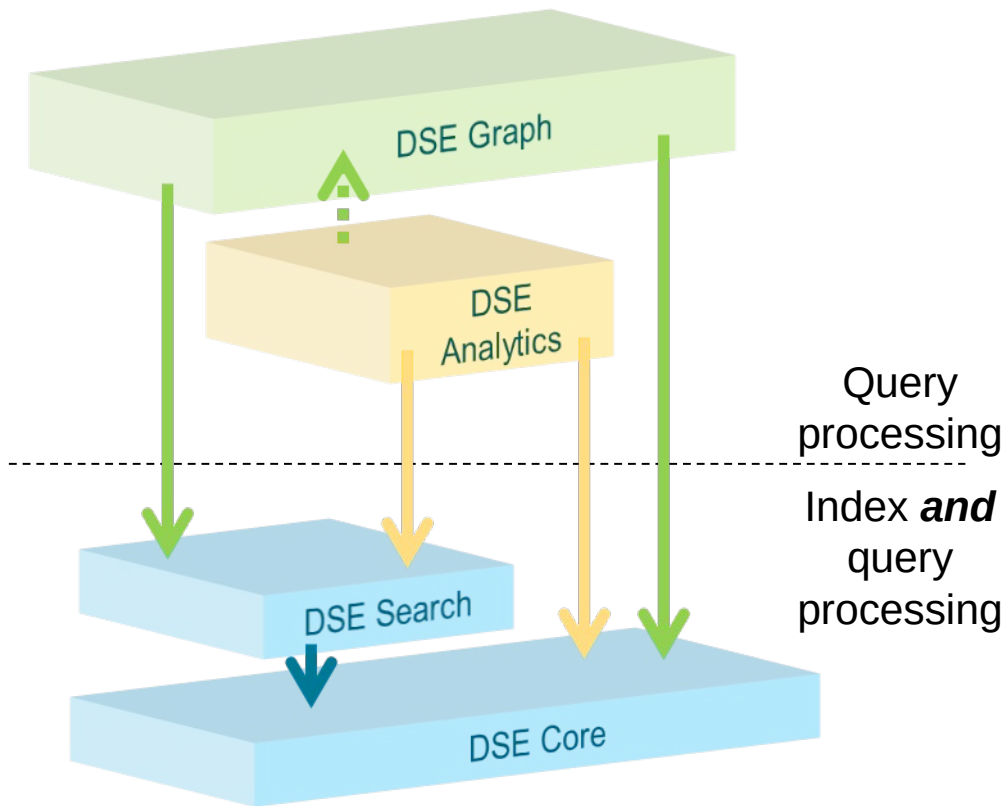


Discussion Module:

Search, Day 2 Review

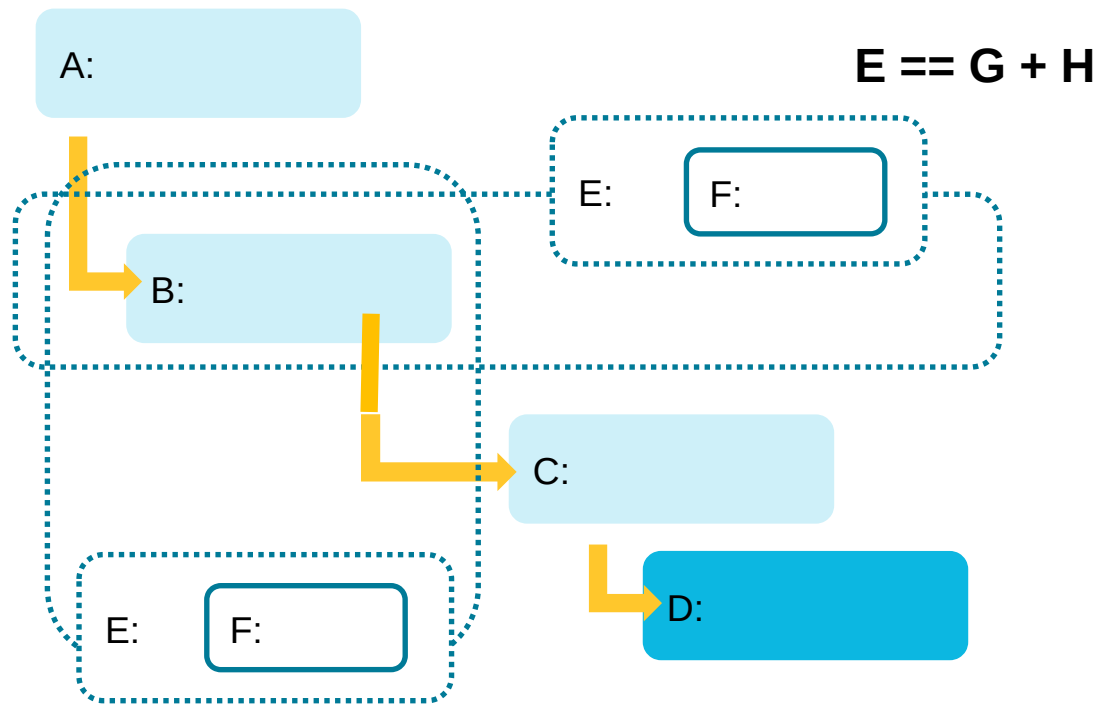
Review of What We Covered on Day 1

The 4 Primary Functional Areas to DSE



- Only 2 of the 4 primary functional areas to DSE provide *indexing technology*
 - DSE Core, *primarily* hash
 - DSE Search, Tf-Idf (bitmap)
- All 4 primary functional areas provide query processing
 - Differentiated query predicates, or functionality
 - Specific performance abilities

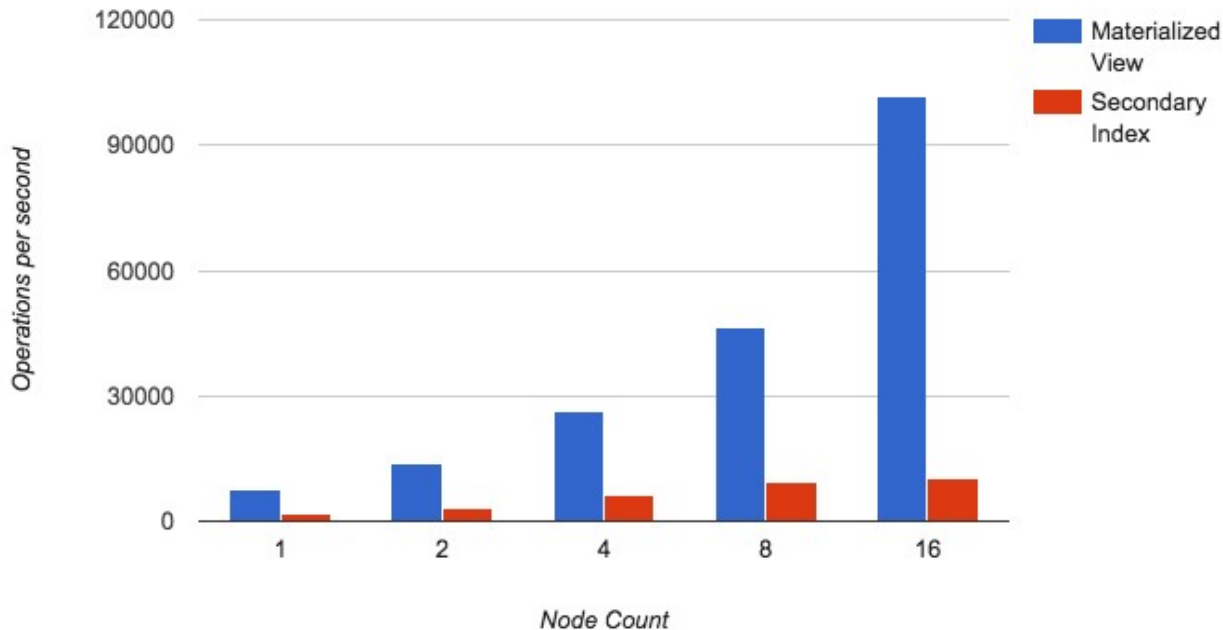
Discussion Lab: (One) DSE Object Hierarchy



- Replication Strategy
- Keyspace
- Clustering Column
- Node (host)
- Table
- Cluster
- Primary Key
- Replication Factor
- Rack
- Data center
- Partitioning Key

And answer the questions on the Notes Page-

Performance: Mat.View versus Secondary Indexes



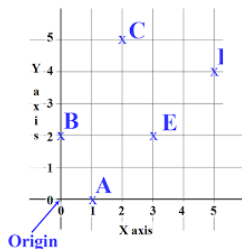
“The performance difference is dramatic even for small clusters, but even more important we see that indexed performance levels off when doubling from 8 to 16 nodes in the (AWS m3.xl) cluster, as the scatter/gather overhead starts to become significant.

Source:

<https://www.datastax.com/dev/blog/materialized-view-performance-in-cassandra-3-x>

DSE Search Index

- Sorting
- Facet (Group by)
- Wildcard (Text)
- Boosting
- Proximity
- (Other)



Integer, Double, String, ...

scalars

2

spatial

GeoSpatial

text

1

Sounds Like
Case Insensitive
Stemming
(Other)

3

Distance from point

Contained (other); Box,
rectangle, polygon

(Time Series)



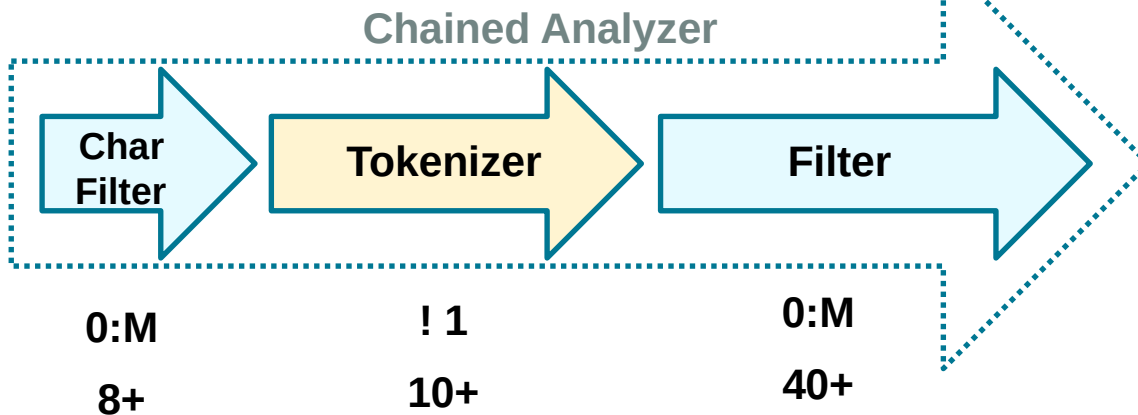
DSE Search: the Analyzer

Analysis Phase: Index|
Query|(Both)




Document
(Column Value)


Document
(Column Value)




	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7
Term(s) 1	10	0	1	0	0	0	0
Term(s) 2	0	2	0	0	0	18	0
Term(s) 3	0	0	0	0	0	0	0
Term(s) 4	6	0	0	4	6	0	0
Term(s) 5	0	0	0	0	0	0	0
Term(s) 6	0	0	1	0	0	1	0
Term(s) 7	0	1	8	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0

Document Vector

Word Vector (Passage Vector)

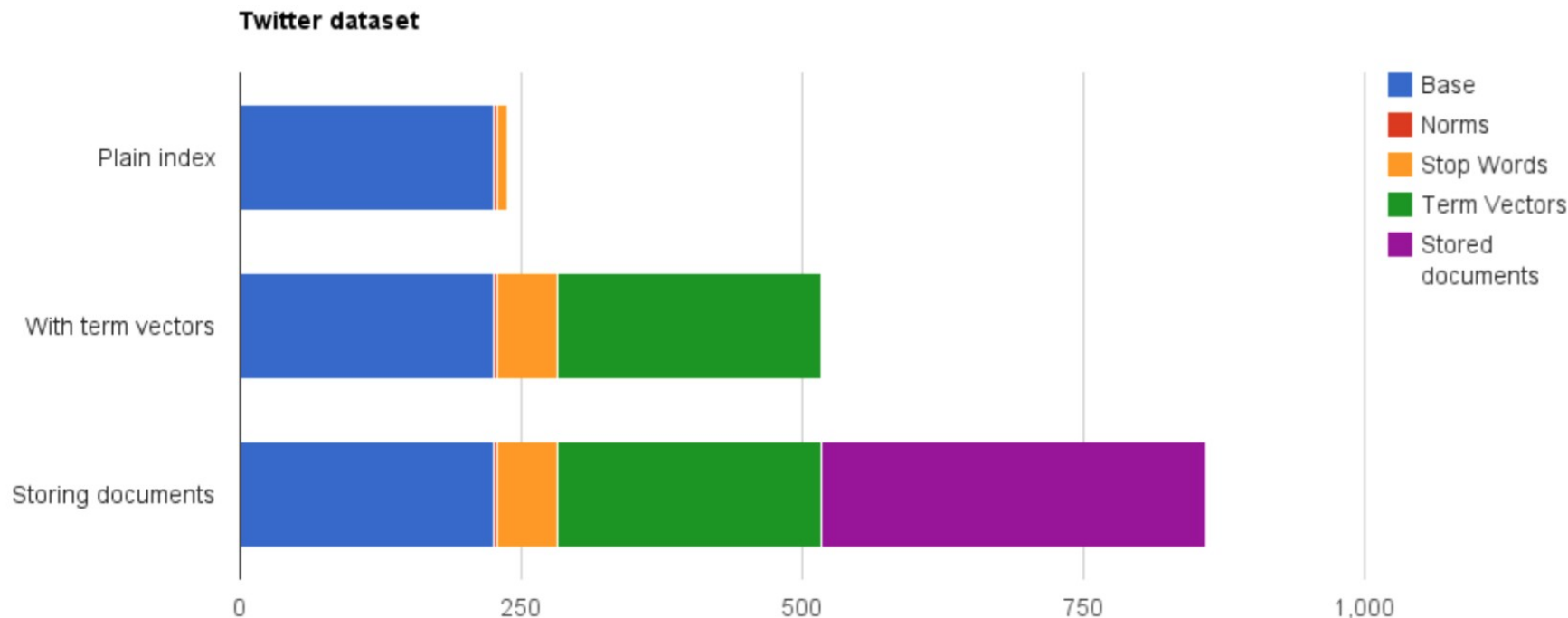


	Document 1	Document 2	Document 3	Document 4	Document 5	Document 6	Document 7
Term(s) 1	10	0	1	0	0	0	0
Term(s) 2	0	2	0	0	0	18	0
Term(s) 3	0	0	0	0	0	0	0
Term(s) 4	6	0	0	4	6	0	0
Term(s) 5	0	0	0	0	0	0	0
Term(s) 6	0	0	1	0	0	1	0
Term(s) 7	0	1	8	0	0	0	0
Term(s) 8	0	0	0	0	0	3	0

Document Vector

Word Vector (Passage Vector)

Apache Solr, Impact on Disk Space, termVectors, more ..



Source: Slide 36, <https://www.slideshare.net/DataStax/solr47-f-inal?https://data.world/datasets/twitter>

Example to test with .. page 6 of 12

```
ALTER SEARCH INDEX SCHEMA ON t1
  ADD types.fieldType[@name='TextField18',
    @class='solr.TextField'] with
    '{"analyzer":{"tokenizer":
      {"class":"solr.StandardTokenizerFactory"},
      "filter":{"class":"solr.BeiderMorseFilterFactory",
        "nameType":"GENERIC", "ruleType":"APPROX",
        "concat":"true", "languageSet":"auto"}}}';
ALTER SEARCH INDEX SCHEMA ON t1
  ADD fields.field[@name='col4',
    @type='TextField18',
    @indexed='true',
    @multiValued='false',
    @stored='true'];
RELOAD SEARCH INDEX ON t1;
REBUILD SEARCH INDEX ON t1;
```

col4



Solr Admin UI: Core -> Analysis -> Column



Dashboard

Logging

Core Admin

Java Properties

Thread Dump

ks_7445.t1

Overview

Analysis

Config

Documents

Files

Ping

Plugins / Stats

Query

Schema

Schema Browser

Segments info

Field Value (Index)
Dave

Field Value (Query)

Analyse Fieldname / FieldType: col3

☒ Verbose Output

[Analyse Values](#)

ST	text	Dave
raw_bytes	[44 61 76 65]	
start	0	
end	4	
positionLength	1	
type	<ALPHANUM>	
position	1	

SE	text	Dave	David	Davie	Dava	Roonie
raw_bytes	[44 61 76 65]	[44 61 76 69 64]	[44 61 76 69 65]	[44 61 76 61]	[52 6f 6f 6e 69 65]	
start	0	0	0	0	0	
end	4	4	4	4	4	
positionLength	1	1	1	1	1	
type	<ALPHANUM>	SYNONYM	SYNONYM	SYNONYM	SYNONYM	
position	1	1	1	1	2	

[Documentation](#) [Issue Tracker](#) [IR](#)

- Debug existing analyzer
 - (Have to manually decompose: Technique)
- Index | Query | (Both)
- Displays (resource file data)

Solr Admin UI: Core -> Query

- \$\$\$
- Adhoc queries
- fl, *.score
- debugQuery (SET EXPLAIN)
- *Much more*

Request-Handler (qt)

/select

— common —

q

col4:starbuku

fq

sort

start, rows

0 10

fl

*, score

df

json.facet

Raw Query Parameters

key1=val1&key2=val2

wt

json

☒ indent

☒ debugQuery

☐ dismax

☐ edismax

http://localhost:8983/solr/ks_7445.t1/select?

```
{
  "responseHeader": {
    "status": 0,
    "QTime": 6
  },
  "response": {
    "numFound": 1,
    "start": 0,
    "maxScore": 4.2663274,
    "docs": [
      {
        "_uniqueKey": "[\"ddd\", \"ddd\"]",
        "col8": "ddd",
        "col9": "ddd",
        "col6": "ddd",
        "col7": "ddd",
        "col4": "Starbucks",
        "col5": "ddd",
        "col2": "ddd",
        "col3": "Mary",
        "col0": "ddd",
        "col1": "ddd",
        "score": 4.2663274
      }
    ]
  },
  "debug": {
    "rawquerystring": "col4:starbuku",
    "querystring": "col4:starbuku",
    "parsedquery": "SynonymQuery(Synonym(co",
    "parsedquery_toString": "Synonym(col4:s",
    "explain": {
      "[\"ddd\", \"ddd\"]": "\n4.2663274 = w",
    },
    "QParser": "LuceneQParser",
  }
}
```

Field Type, Field, Attributes: Covered

StrField ☐

TextField ☐

TrieDoubleField ☐



- ☐ indexed=true
- ☐ indexed=false
- ☐ stored=t|f
- ☐ docValues=t|f
- ☐ multiValued=true
- ☐ multiValued=false
- ☐ default=*str*
- ☐ precisionStep=n
- ☐ positionIncrementGap=n
- ☐ termVectors=t|f
- ☐ termPositions=t|f
- ☐ termOffsets=t|f
- ☐ omitNorms=t|f

Field Type, Field, Attributes: Covered

StrField ☐

TextField ☐

TrieDoubleField ☐

Word Proximity ☐

docValues ☐



☐ **KeywordTokenizer**

☐ **StandardTokenizer**

☐ **docValues**

CQL SELECT: (Core, other)

SELECT ..
FROM ..
WHERE ..
GROUP BY ..
ORDER BY .. ;



+ Spark

2.2.0.14



(JOINS)



1.2.1

HIVE

- LIMIT | PARTITION LIMIT
- UDFs, UDAs, (+ native funcs & aggs)
(Core Yes, Core + Search, No)
- Paging (Not graph; Core and Search Yes)

Query: Solr Server Side “Function Queries”

```
SELECT col1, col3, col4 FROM t10
WHERE
solr_query = '{ "q":":*:*",
  "sort": "sum(col3, col4) asc" }';
```

col1	col3	col4
aaa	1	2
ccc	1	5
ddd	1	6
bbb	3	4

- See https://lucene.apache.org/solr/guide/6_6/function-queries.html
- 20 or more by count, not all are relevant
- Used in predicates, value can not be returned to the client

```
SELECT col1, col6 FROM t10
WHERE solr_query = '{ "q" :
  "{! q.op=AND}col6:(Mouse Mule)" }';
```

```
col1 | col6
-----+-----
(0 rows)
```

```
...
'{ "q" : "{! q.op=OR}col6:(Mouse Mule)" }';
```

```
col1 | col6
-----+-----
ccc | Dog Mule Cat
bbb | Dog Mouse Cat
```

Standard Query Parser: Parameters, q.op

Data is :

```
'aaa', 'Mary' , 'Dog Cat Bird'
'bbb', 'Harry', 'Dog Mouse Cat'
'ccc', 'Dave' , 'Dog Mule Cat'
'ddd', 'David', 'Cat Dog Bird'
```

Query: Facet, “distributed pivot facet”, aka, decision tree

```
SELECT col1 FROM t10  
WHERE solr_query='{ "q": "col1:*",  
  "facet": { "pivot": "col3,col4",  
    "limit": "-1" } }';
```

See formatted query result on notes page.



```
-----  
"col3,col4" :  
  "field" : "col3",  
  "value" : 1,  
  "count" : 3,  
  "pivot" :  
    -----  
    "field" : "col4",  
    "value" : 2,  
    "count" : 1  
    -----  
    "field" : "col4",  
    "value" : 5,  
    "count" : 1  
    -----  
    "field" : "col4",  
    "value" : 6,  
    "count" : 1  
    -----  
  "field" : "col3",  
  "value" : 3,  
  "count" : 1,  
  "pivot" :  
    "field" : "col4",  
    "value" : 4,  
    "count" : 1  
-----
```

PC Mouse Feature

- ☐ Wireless
- ☐ Optical
- ☐ Notebook
- ☐ Wheel

Mouse Interface

- ☐ USB
- ☐ Bluetooth
- ☐ RF Wireless
- ☐ PS/2
- ☐ Infrared

Mouse Platform Support

- ☐ Mac
- ☐ PC
- ☐ Linux



End of Module: